**FINANCIAL MANAGEMENT**

Suggested Answers
November-December 2025

**Answer to the Question# 1(a):**

**Rights Issue Calculations**

**Let us assume a current market price of BDT 90 per share**

**Theoretical Ex-Rights Price (TERP):**
A 1-for-4 rights issue means an investor can buy 1 new share for every 4 they currently own.

Value of 4 existing shares @ BDT 90 = 4 X 85 = BDT 360
Cost of 1 new share @ BDT 70 = 1 times 70 = BDT 70
Total value of 5 shares = 360 + 70 = BDT 430
TERP = 430 /5 = BDT 86 per share.

**Value of one right:**
The value of a right is the difference between the TERP and the issue price of the new share.
Value of right = TERP - Issue Price = 86.00 - 70.00 = BDT 16.00

**Answer to the Question# 1(b):**

**EPS and Financial Gearing (Year ended 31 Dec 2025)**

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**Step 1: Forecast PBIT (2025)**

$$2{,}500 + 120 = 2.620\ million$$

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**Option 1: Rights Issue Financing**
**Interest expense**

- Existing bonds:

$$4{,}000 \times 8\% = 320$$

**Profit after interest (ignore tax – EPS comparison basis)**

$$2{,}620 - 320 = \boxed{2{,}300}$$

**Number of shares**

$$200m + 50m = \boxed{250m}$$

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**EPS (Rights Issue)**

$$EPS = \frac{2{,}300}{250} = 9.20$$

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**Financial Gearing (Rights Issue)**
**Market value of equity (TERP basis):**
$250m \times 86 = 21{,}500$

**Market value of debt (existing only):**
4,000

$$\text{Gearing} = \frac{4{,}000}{21{,}500} = \boxed{18.6\%}$$

**Option 2: Bond Issue Financing**
**New bond interest**
$800 \times 7\% = 56$

**Total interest**
$320 + 56 = 376$

**Profit after interest**
$2{,}620 - 376 = \boxed{2{,}244}$

**Number of shares**
$\boxed{200m}$

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**EPS (Bond Issue)**

$$EPS = \frac{2{,}244}{200} = 11.22$$

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**Financial Gearing (Bond Issue)**
**Market value of debt:**
$4{,}000 + 768 = 4{,}768$

**Market value of equity:**
$200m \times 90 = 18{,}000$
Gearing $= \frac{4{,}768}{18{,}000} = \boxed{26.5\%}$

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**(c) Advice to the Board**

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**EPS comparison**

| Option | EPS (BDT) |
|---|---|
| Rights issue | **9.20** |
| Bond issue | **11.22** |

**Bond issue produces significantly higher EPS** due to:

- No dilution of shares
- Fixed interest cost lower than earnings generated

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**Financial gearing comparison**

| Option | Gearing |
|---|---|
| Rights issue | **18.6%** |
| Bond issue | **26.5%** |
| Industry average | **40%** |

- Both options keep gearing **below industry average**
- Bond issue improves gearing efficiency without excessive risk

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**Risk considerations**
**Rights issue**
✓ Lower financial risk
✗ Dilution of EPS
✗ Possible dilution of ownership if shareholders do not subscribe
✗ May signal weak cash-flow confidence
**Bond issue**

✓ Higher EPS
✓ No dilution of control
✓ Gearing still conservative
✕ Increased financial risk (fixed interest obligations)

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**Shareholder control**

- **Rights issue** risks:
  - Ownership dilution
  - Under-subscription
- **Bond issue**:
  - Preserves control
  - No voting rights attached

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**Final recommendation**

**Meghna should choose the BOND ISSUE**, because:

- EPS is **significantly higher**
- Gearing remains **below industry norms**
- Financial risk is acceptable given strong PBIT
- Shareholder control is preserved
- Cost of debt (7%) is lower than project return

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**Overall conclusion**

The **bond issue dominates the rights issue financially and strategically**, making it the **preferred financing option**.

**Answer to the Question# 2(a):**

**Mack Ltd**

**(i) Forecasted statement of profit or loss for the year ended 31 December 2024**

| | Tk.'000 | Tk.'000 |
|---|---|---|
| Revenue [Tk. 240 million x 1.25] | | 300,000 |
| Cost of sales [70%] | | 210,000 |
| Gross profit | | 90,000 |
| Expenses | 54,500 | |
| Depreciation of building [5% of Tk. 20 million] | 1,000 | |
| Depreciation of plant and machinery [20% of Tk. (25 million + 57 million] | 16,400 | |
| | | 71,900 |
| Profit before taxation | | 18,100 |
| Taxation (25%) | | 4,525 |
| Profit after taxation | | 13,575 |
| Dividend (50%) | | 6,788 |
| Retained profit | | 6,788 |

**Mack Ltd**

**(ii) Forecasted balance sheet as on 31 December 2024**

| | Tk.'000 | Tk.'000 | Tk.'000 |
|---|---|---|---|
| Non-current assets | | | |
| Freehold buildings at cost | | 25,000 | |
| Less: accumulated depreciation | | 6,000 | 19,000 |
| | | | |
| Plant and machinery at cost | | 94,900 | |
| Less: accumulated depreciation [12.9 million + 16.4 million) | | 29,300 | 65,600 |

| | | | |
|---|---|---|---|
| | | | 84,600 |
| Current assets | | | |
| Inventory [39 million - 18 million] | | 21,000 | |
| Trade receivables (N-1) | | 75,000 | |
| | | 96,000 | |
| Less payables: amounts falling due within one year | | | |
| Bank overdraft [N-4] | 30,217 | | |
| Trade payables [N-2] | 24,000 | 54,217 | 41,783 |
| | | | 126,383 |
| Long term loan [N-3] | | | 31,596 |
| | | | 94,788 |
| | | | |
| Share capital and reserves | | | |
| Ordinary shares @ Tk. 10 each | | | 25,000 |
| Accumulated profits | | | 69,788 |
| | | | 94,788 |
| | | | - |

Working notes:

N-1: Trade receivables

Average debt collection period in year to 31 December 2023 = (20 / 240) x 12 months = 1 month
Average debt collection period in year to 31 December 2024 = 3 times one month = 3 months
Trade receivables as at 31 December 2024 = (3/12) x Tk. 300 million = Tk. 75 million

N-2: Trade payables

Purchases in the year = closing inventory + cost of sales – opening inventory.
Purchases are therefore (given Tk. 18 million reduction in inventory levels) 21m + 210m – 39m = Tk. 192 million.
Trade payables at 31 December 2024 = (1.5 /12) x Tk. 192 million = Tk. 24 million

N-3: Long term loan

| | Tk.'000 |
|---|---|
| Share capital | 25,000 |
| Reserves as at 31 December 2023 | 63,000 |
| Retained profit 2024 | 6,788 |
| Share capital and reserves at 31 December 2024 | 94,788 |
| Loan at 31 December 2024 (one-third of Tk. 94,788) | 31,596 |
| Original loan | 57,000 |
| Loan repaid in the year | 25,404 |

N-4: Cash

| | Tk.'000 | Tk.'000 |
|---|---|---|
| Operating profit | | 18,100 |
| Add back depreciation (Tk. 1 million + Tk. 16.4 million) | | 17,400 |
| | | 35,500 |
| Reduction in inventory | 18,000 | |
| Increase in receivables (Tk. 75 million - Tk. 20 million) | (55,000) | |
| Increase in trade payables (Tk. 24 million - Tk. 16.5 million) | 7,500 | |
| | | (29,500) |
| Cash flow from operation | | 6,000 |
| Tax paid | | (4,525) |
| Dividend paid | | (6,788) |
| Machinery purchase | | (57,000) |
| Loan raised | | 57,000 |
| Loan repaid | | (25,404) |
| Cash flows in the year | | (30,717) |
| Cash at the start of the year | | 500 |

| | |
|---|---|
| Cash at the end of the year | (30,217) |

**Answer to the Question# 2(b):**

Liquidity is expected to deteriorate. The company is forecasting a bank overdraft of Tk. 30.2 million at the end of 31 December 2024, compared to a positive cash balance the previous year. The company will have to ensure that bank overdraft facilities are available if it goes ahead with its plan to increase sales.

The company will also have some long-term debt capital, having been ungeared in the year to 31 December 2023.

On the basis of the forecasts, it is questionable whether there is any financial benefit to be obtained from expanding sales, and the company management should review its plans urgently, before implementing them.

**Answer to the Question# 3(a):**

**Hedging the USD 2,500,000 Payment**

Meghna needs to buy USD, so it will use the bank's selling rate (the higher rate).

1. Forward Exchange Contract:

Meghna will agree to buy USD 2,500,000 in three months at the forward rate of BDT 111.40 per USD.
Cost in BDT = BDT 2,500,000 X 111.40 = BDT 278,500,000

2. Money Market Hedge:

This involves creating a synthetic forward rate. To make a USD payment in 3 months, Meghna must have a USD deposit that matures to USD 2.5m on that date.

- Step 1: Calculate USD to be deposited now.
  - Meghna will deposit into a USD account. The bank's USD deposit rate is 4.50% per annum.
  - Three-month rate = 4.50% X 3/ 12 = 1.125%
  - USD to deposit =2,500,000} /{1 + 0.01125} = USD 2,472,188

- Step 2: Buy the required USD on the spot market now.
  - Meghna buys USD at the spot selling rate of BDT 110.65 per USD.
  - BDT cost today = 2,472,188X 110.65 = BDT 273,547,602.

- Step 3: Borrow BDT now to fund the spot transaction.
  - Meghna borrows BDT from the bank. The bank's BDT borrowing rate is 7.50% per annum.
  - Three-month rate = 7.50% X 3/12 = 1.875%
  - Total BDT to repay in 3 months = 273,547,602X (1 + 0.01875) = BDT 278,676,620

Summary for USD Payment:
- Forward Contract Cost: BDT 278,500,000
- Money Market Hedge Cost: BDT 278,676,620

**Answer to the Question# 3(b):**

**Hedging the EUR 1,000,000 Receipt**

Meghna needs to sell EUR, so it will use the bank's buying rate (the lower rate).

1. Forward Exchange Contract:

Meghna will agree to sell EUR 1,000,000 in three months at the forward rate of BDT 124.50 per EUR.

Receipt in BDT = 1,000,000 X 124.50 = BDT 124,500,000

2. Money Market Hedge:

To hedge a EUR receipt, Meghna will borrow EUR now, convert to BDT, and deposit the BDT.

- Step 1: Borrow EUR now.
  - Meghna borrows from a EUR bank. The EUR borrowing rate is 4.00% per annum.
  - Three-month rate = 4.00% X 3/12 = 1.00%
  - The EUR receipt in 3 months will be used to repay this loan.
  - EUR to borrow now = 1,000,000 / {1 + 0.01} = 990,099
- Step 2: Convert the EUR to BDT on the spot market now.
  - Meghna sells EUR at the spot buying rate of BDT 125.10 per EUR.
  - BDT received today = 990,099 X 125.10 = BDT 123,861,386

- Step 3: Deposit the BDT now.
  - Meghna deposits the BDT. The BDT deposit rate is 6.00% per annum.
  - Three-month rate = 6.00% X 3/12 = 1.50%
  - Total BDT received in 3 months = 123,861,386 X (1 + 0.015) = BDT 125,719,307

Summary for EUR Receipt:
- Forward Contract Receipt: BDT 124,500,000
- Money Market Hedge Receipt: BDT 125,719,307

**Answer to the Question# 3(c):**

**Advice:**

- For the USD 2.5m payment: Meghna should choose the option with the lower BDT cost. The forward contract costs BDT 278,500,000, while the money market hedge costs BDT 278,644,594. Therefore, Meghna should use the forward contract.

- For the EUR 1.0m receipt: Meghna should choose the option with the higher BDT receipt. The money market hedge yields BDT 125,719,307, while the forward contract yields BDT 124,500,000. Therefore, Meghna should use the money market hedge.

**Answer to the Question# 4(a):**

Exposure and Hedge Coverage:
Total exposure: €1,500,000
Hedge ratio: 80% = €1,200,000
Futures contract size: €125,000
Number of contracts: €1,200,000 / €125,000 = 9.6
= 10 contracts (rounded)

Futures Gain:
Futures bought at: USD 0.9205
Futures closed at: USD 0.9455
Gain per €: 0.9455 - 0.9205 = 0.0250
Total gain: 10 x €125,000 x 0.0250 = USD 31,250

Spot Payment:
Spot rate on 30 Sept: USD 0.9428
Total cost at spot: €1,500,000 x 0.9428 = USD 1,414,200

Net Cost After Hedge:
Net cost = Spot payment - Futures gain

= 1,414,200 - 31,250
= USD 1,382,950

Effective Exchange Rate:
Effective rate = Net cost / Total euros
1,382,950 / 1,500,000 = USD 0.92197 per €

**Answer to the Question# 4(b):**

Meditech's hedge was effective in reducing currency risk but not fully comprehensive:

**Partial coverage**: Only 80% of the €1,500,000 exposure was hedged, leaving €300,000 unhedged. Due to euro appreciation from USD 0.9180 to 0.9428, the unhedged portion incurred an additional cost of: €300,000 x (0.9428 - 0.9180) = **USD 7,440**.

**Basis movement**: The basis widened slightly from +0.0025 to +0.0027, meaning futures prices rose more than spot. This favored Meditech's long futures position, increasing the hedge gain and slightly improving the effective exchange rate.

**Instrument selection**: Futures provided liquidity and daily mark-to-market visibility, but introduced basis risk and margin requirements. For a fixed-date payment, forwards could have eliminated basis risk. Options would have preserved upside if the euro had weakened, but at a premium cost.

**Strategic alignment**: Despite partial exposure, the hedge reduced volatility and protected budget certainty for a critical Q4 product launch, supporting investor confidence and operational timelines.

**Answer to the Question# 4(c):**

Basis risk is the risk that the futures price and spot rate will not converge as expected at settlement, potentially distorting hedge outcomes. It arises from interest rate differentials, market expectations, and liquidity conditions. In this case:

- **Basis at inception (1 August)** = 0.9205 - 0.9180 = **+0.0025**
- **Basis at settlement (30 September)** = 0.9455 - 0.9428 = **+0.0027**

The basis widened slightly, which benefited Meditech's long futures position by increasing the hedge gain. While favorable here, basis risk introduces uncertainty and can reduce hedge precision especially when coverage is partial or timing mismatches occur.

**Answer to the Question# 5(a)(i):**

NPV calculation:

| Particulars | Year 0 | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 | Year 6 |
|---|---|---|---|---|---|---|---|
| Revenue | - | 295.00 | 295.00 | 295.00 | 295.00 | 295.00 | 295.00 |
| Variable Costs | - | 40.00 | 40.00 | 40.00 | 40.00 | 40.00 | 40.00 |
| Fixed Costs | - | 20.00 | 20.00 | 20.00 | 20.00 | 20.00 | 20.00 |
| Insurance | - | 5.00 | 5.00 | 5.00 | 5.00 | 5.00 | 5.00 |
| Maintenance | - | - | 8.00 | 8.00 | 8.00 | 8.00 | 8.00 |
| Energy (6% escalation) | - | 12.00 | 12.72 | 13.48 | 14.29 | 15.15 | 16.06 |
| Staffing | - | 9.00 | 9.00 | 9.00 | 9.00 | 9.00 | 9.00 |
| Opportunity Cost (Land) | - | 15.00 | 15.00 | 15.00 | 15.00 | 15.00 | 15.00 |
| Total Operating Costs | - | 101.00 | 109.72 | 110.48 | 111.29 | 112.15 | 113.06 |
| Depreciation (BDT 700m) | - | 116.67 | 116.67 | 116.67 | 116.67 | 116.67 | 116.67 |
| EBIT | - | 77.33 | 68.61 | 67.85 | 67.04 | 66.18 | 65.27 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Tax (0% Y1- 3, 30% Y4- 6) | - | - | - | - | 20.11 | 19.85 | 19.58 |
| Net Income | - | 77.33 | 68.61 | 67.85 | 46.93 | 46.33 | 45.69 |
| Add Back Depreciation | - | 116.67 | 116.67 | 116.67 | 116.67 | 116.67 | 116.67 |
| Investment | -700.00 | - | - | - | - | - | - |
| Working capital | -60.00 | - | - | - | - | - | 60.00 |
| Net Cash Flow | -760.00 | 194.00 | 185.28 | 184.52 | 163.60 | 163.00 | 222.36 |
| Discount Factor @13% | 1.0000 | 0.8850 | 0.7831 | 0.6925 | 0.6126 | 0.5428 | 0.4803 |
| Present Value of Cash Flow | -760.00 | 171.68 | 145.10 | 127.88 | 100.34 | 88.47 | 106.80 |
| Net present value | -19.73 | | | | | | |

Notes:
1. Sunk Cost (BDT 25 million pre-operating expenses): Already incurred and non-recoverable. Excluded from NPV as per capital budgeting principles.

2. Incremental Costs (BDT 9 million staffing): Directly attributable to the project and not part of existing operations. Included in annual operating costs.

3. Opportunity Cost (BDT 15 million per year for foregone logistics hub): Represents economic value sacrificed. Treated as a real cash outflow each year.

4. Administrative Overheads (BDT 6 million per year): Not incremental would be incurred regardless of project approval. Excluded from cash flows.

5. WACC = 45% x 10% x (1 - 0.30) + 55% x [7% + 2.18 x 5%] = 3.15% + 10.3% = 13%.

6. Amounts in million BDT.

**Answer to the Question# 5(a)(ii):**

IRR calculation:

Since the NPV at 13% is negative and the margin is small, even a slight reduction in the cost of capital could make the NPV positive. At 12%, the NPV is 19.50 million, calculated using the same cash flow method applied for the 13% case, and the result is close to that at 13%. This confirms that the IRR lies between 12% and 13%. Now, applying interpolation:

IRR = Lower rate + (Positive NPV / (Positive NPV - Negative NPV)) x (Difference in rates)
= 12% + (19.50 / (19.50 - (-19.73))) × (13% -12%)
= 12% + (19.50 / 39.23) x 1%
= 12% + 0.50%
= 12.5%

**Answer to the Question# 5(b):**

Financial and strategically viability:

The Chattogram facility requires an upfront investment of BDT 760 million, above the board's ceiling of 500 million. Financial appraisal shows a slightly negative NPV of -19.73 million and an IRR of about 12.5 percent, just below the 13 percent cost of capital, indicating marginal financial viability. Strategically, however, the project offers clear benefits in ESG alignment, export diversification, and supply chain control, though risks such as currency volatility and regulatory uncertainty remain.

Given these results, the project should not be approved outright but considered conditionally. Adjustments in financing, risk management, and phased implementation could improve feasibility. With modest revenue growth or favorable exchange rates, the project could turn positive and become a viable investment supporting Agrochem's long-term diversification goals.

**Answer to the Question# 5(c):**

| Particulars | **Year 0** | **Year 1** | **Year 2** | **Year 3** | **Year 4** | **Year 5** |
|---|---|---|---|---|---|---|
| Current EBIT (from base case) | - | 77.33 | 68.61 | 67.85 | 67.04 | 66.18 |
| Add: EBIT from 15% revenue increase | - | 38.25 | 38.25 | 38.25 | 38.25 | 38.25 |
| Revised EBIT | - | 115.58 | 106.86 | 106.10 | 105.29 | 104.43 |
| Less: Tax @30% (Y4 -Y5 only) | - | - | - | - | 31.58 | 31.33 |
| Add: Depreciation (unchanged) | - | 116.67 | 116.67 | 116.67 | 116.67 | 116.67 |
| Add: Working Capital Recovery | -60.00 | - | - | - | - | 60.00 |
| Net Cash Flow | - 760.00 | 232.25 | 223.53 | 222.77 | 190.37 | 249.77 |
| Discount Factor @13% | 1.0000 | 0.8850 | 0.7831 | 0.6925 | 0.6126 | 0.5428 |
| Present Value | -760.00 | 205.24 | 175.05 | 154.27 | 116.62 | 135.58 |
| NPV | 26.76 | | | | | |

The revised scenario improves annual cash flows through higher revenue and earlier recovery of working capital. Although Year 6 cash flows are lost, the discounted benefits outweigh the upfront investment. As a result, the NPV is positive at BDT 26.76 million, and the project is financially viable under a five-year horizon.

Before making a final decision, the board should also assess key risks and strategic considerations. These include foreign exchange exposure on the JPY 400 million machinery import, execution risk in scaling up output, and the opportunity cost of foregoing BDT 15 million annually from the alternative logistics hub. At the same time, the project offers strategic benefits such as ESG alignment, export diversification, and greater supply chain control. Regulatory uncertainty and inflationary pressures should also be factored into the final evaluation.

**Answer to the Question# 6(a):**

**Step 1: Understand each project's cash flows**
We have **t0, t1, t2** in Tk.'000.

| Project | t0 | t1 | t2 |
|---|---|---|---|
| A | (15,000) | (2,500) | 32,500 |
| B | (20,000) | (6,500) | 35,000 |
| C | (17,500) | 5,000 | 15,000 |
| D | (25,000) | (2,000) | 40,000 |
| E | (16,000) | (7,500) | 29,500 |

**Discount rate = 10%**

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**NPV formula:**

$$NPV = CF_0 + \frac{CF_1}{(1.1)} + \frac{CF_2}{(1.1)^2}$$

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**Project A:**

$$NPV_A = -15 + \frac{-2.5}{1.1} + \frac{32.5}{1.1^2}$$
$$= -15 - 2.2727 + \frac{32.5}{1.21}$$
$$= -15 - 2.2727 + 26.8595$$

$= 9.5868$ (Tk.'000)→ Tk. 9,587

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**Project B:**

$$NPV_B = -20 + \frac{-6.5}{1.1} + \frac{35}{1.21}$$
$$= -20 - 5.9091 + 28.9256$$
$$= 3.0165 \text{ (Tk.'000)} \rightarrow \text{Tk. } 3{,}017$$

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**Project C:**

$$NPV_C = -17.5 + \frac{5}{1.1} + \frac{15}{1.21}$$
$$= -17.5 + 4.5455 + 12.3967$$
$$= -0.5578 \text{ (Tk.'000)} \rightarrow \text{Tk. } -558$$

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**Project D:**

$$NPV_D = -25 + \frac{-2}{1.1} + \frac{40}{1.21}$$
$$= -25 - 1.8182 + 33.0579$$
$$= 6.2397 \text{ (Tk.'000)} \rightarrow \text{Tk. } 6{,}240$$

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**Project E:**

$$NPV_E = -16 + \frac{-7.5}{1.1} + \frac{29.5}{1.21}$$
$$= -16 - 6.8182 + 24.3802$$
$$= 1.5620 \text{ (Tk.'000)} \rightarrow \text{Tk. } 1{,}562$$

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**Ranking by NPV (in Tk.'000):**
A: 9.587
D: 6.240
B: 3.017
E: 1.562
C: –0.558

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**a) Perfect capital market, independent divisible projects**
Accept all with **NPV > 0**, reject C.
So accept **A, B, D, E**.

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**b) Capital rationing at t0: only Tk. 50,000 available**
**Divisible** projects → choose by **Profitability Index (PI)**.

$$PI = \frac{\text{PV of future CFs}}{|\text{Initial outlay at t0}|}$$

We need **t0 outlay** only? Wait: t0 cash flows are initial investment **negative**. Additional outlays at t1 should also be considered as part of the initial investment for rationing **at t0 only**?
In capital rationing at t0, we can borrow/lend freely at t1 (perfect capital market in subsequent periods) so the constraint is only t0 outlay?
Usually: Capital rationing only at t0 → we rank by **NPV per Tk invested at t0**.
But t1 outlay is **also negative** for some projects; but t1 is not constrained here. So PI could be misleading if t1 outlays differ.
Better method: Since t1 is not constrained, we can bring t1 outlay to t0 at 10% cost, so total present value of costs at t0:
Total PV of costs at t0 = $CF_0 + \frac{CF_1}{1.1}$ for CF1 negative only.

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Let's find **NPV per Tk of PV of outlays at t0** (t0 outlay only? or t0+t1 outlays discounted to t0)?
But in some books, if only t0 rationing, use t0 outlay only for PI. Let's follow that: use **t0 outlay** as denominator (in Tk.'000):

| Project | NPV | t0 outlay | PI = NPV / t0 outlay |
|---|---|---|---|
| A | 9.587 | 15 | 0.6391 |
| D | 6.240 | 25 | 0.2496 |
| B | 3.017 | 20 | 0.1509 |
| E | 1.562 | 16 | 0.0976 |
| C | –0.558 | 17.5 | (negative) |

Rank by PI: **A (0.639), D (0.250), B (0.151), E (0.098)**.
Budget 50,000:
Choose A → 15 used, 35 left.
Choose D → 25 used, 10 left.
Next B needs 20, not possible.
E needs 16, not possible (only 10 left).
C is negative.
We could take **A fully, D fully, and part of B** if divisible.
Check part of B:
With 10 left after A+D (15+25=40 used, 10 left), invest in 10/20 = 0.5 of B.
NPV from A = 9.587
NPV from D = 6.240
NPV from 0.5 of B = 0.5 × 3.017 = 1.5085
Total NPV = 9.587 + 6.240 + 1.5085 = **17.3355 (Tk.'000)**
Check if mixing differently yields higher NPV:
What if A fully (15), B fully (20) → 35 used, 15 left → can take part of D: 15/25 = 0.6 of D.
NPV: A 9.587, B 3.017, 0.6×D = 0.6×6.240 = 3.744 → Total = 16.348 → less than 17.336.
So previous is better: A, D, 0.5B.
We could test E instead of part B: E's PI lower than B's, so no.
So best: **A, D, 0.5B**.

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**c) Projects A and B are mutually exclusive**
With capital rationing 50,000 t0, divisible.
We now compare top combinations without taking both A and B together.
Option 1: Take A and not B:
Rank: A (PI 0.639), D (0.250), E (0.098), C negative.
A (15) + D (25) = 40, leftover 10 → cannot take E fully (needs 16), can take part of E?
But PI of E is less than B's (but B not allowed if A taken). Actually B is excluded here.
So after A+D, leftover 10, choose part of E: 10/16 = 0.625 of E → NPV 0.625×1.562=0.976.
Total NPV = 9.587 + 6.240 + 0.976 = 16.803
Option 2: Take B and not A:
B (20), D (25) total 45, leftover 5 → part of E (5/16=0.3125) → NPV 0.488.
Total NPV = 3.017 + 6.240 + 0.488 = 9.745
Option 3: Skip both A and B? Then pick D (25), E (16) = 41, leftover 9 → can take part of C? C negative, so no.
Actually leftover 9 unused? So just D+E fully:
NPV = 6.240 + 1.562 = 7.802
Clearly Option 1 (A not B) is best: 16.803 vs 9.745 vs 7.802.
Thus choose **A, D, and 0.625 of E**.

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**d) Indivisible projects, capital rationing at t0 (50,000)**
Try combinations within budget:
Combinations (t0 outlay, NPV in '000):

1. A+B = 15+20=35, NPV=9.587+3.017=12.604
2. A+D = 15+25=40, NPV=9.587+6.240=15.827
3. A+E = 15+16=31, NPV=9.587+1.562=11.149
4. B+D = 20+25=45, NPV=3.017+6.240=9.257
5. B+E = 20+16=36, NPV=3.017+1.562=4.579
6. D+E = 25+16=41, NPV=6.240+1.562=7.802
7. A alone = 15, NPV=9.587
8. D alone = 25, NPV=6.240
9. B+E+C? C negative, lowers NPV.
10. A+B+E = 51 > 50 not allowed.
11. A+D+E = 56 > 50 not allowed.

Check triple possible within 50:
A+B+? C? outlay 52.5 >50.
B+D+E=61>50.
A+E+C=48.5 (NPV=9.587+1.562-0.558=10.591) less than A+D (15.827).
Best so far: A+D (40k outlay, NPV=15.827).
Can we do three projects? B+E+C=53.5>50, so no.
What about B+E=4.579 worse.
D+E=7.802 worse.
**Thus best indivisible:** Choose **A and D** (uses 40k, leftover 10k idle).
Max NPV = **Tk. 15,827**.

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**e) Capital rationing at t1 only (Tk. 15,000 available), divisible, independent**
Now constraints at t1 outlays: Some projects require t1 outflow (negative CF1), some have t1 inflow.
t1 cash flows ('000):
A: –2.5
B: –6.5
C: +5
D: –2
E: –7.5
We can use t1 inflows from some to fund t1 outflows of others.
We must choose projects to maximize total NPV subject to:
Sum of t1 outflows ≤ 15,000 (t1 inflows reduce net requirement).
Better: Net t1 outlay = sum over chosen projects of CF1. Since CF1 can be positive (inflow), we can have more projects with negative CF1 as long as total CF1 from chosen projects ≥ –15 (because constraint is only 15 available from external finance, meaning we can cover net outflow up to 15? Actually: available 15 means we can pay up to 15 in outflows; inflows add to availability.)
Let's find **NPV per Tk of t1 outlay (if CF1 negative)** or treat inflows as reducing need.
A simpler approach: Include all projects, calculate net CF1:
If select all: CF1 total = –2.5 – 6.5 + 5 – 2 – 7.5 = –13.5 → needs 13.5 external, less than 15, so all can be chosen if divisible? But C has negative NPV, so exclude C.
If exclude C: CF1 = –2.5 – 6.5 – 2 – 7.5 = –18.5 → need 18.5 external > 15 available. So must drop some.
Rank projects by NPV per Tk of t1 outlay:
NPV / |CF1| for CF1 negative, but CF1 positive (C) is bad NPV anyway.
PI on t1 outlay:
A: 9.587 / 2.5 = 3.8348
B: 3.017 / 6.5 = 0.4642
D: 6.240 / 2 = 3.12
E: 1.562 / 7.5 = 0.2083
C: negative, ignore.
Rank: A, D, B, E.
Now allocate t1 funds 15:
Take A fully → t1 use 2.5, left 12.5.
Take D fully → t1 use 2, left 10.5.
Take B fully → t1 use 6.5, left 4.
Take E fully → needs 7.5, but only 4 left → take 4/7.5 = 0.533 of E.
NPV = 9.587 + 6.240 + 3.017 + 0.533×1.562
= 18.844 + 0.833 ≈ 19.677 (Tk.'000)

Check if skipping B to take E fully: A+D+E use t1 = 2.5+2+7.5=12, leftover 3 → could take part of B? No leftover after that. Actually:
A+D+E: NPV=9.587+6.240+1.562=17.389, leftover t1 capacity 3 unused.
Less than 19.677, so better to take B and part of E.
So choose **A, D, B, 0.533 of E**.

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**Final Answers:**
a) Accept A, B, D, E (all positive NPV except C).
b) Accept A fully, D fully, 0.5 of B (uses t0: 15+25+10=50), NPV=17.3355.
c) Accept A fully, D fully, 0.625 of E (since A and B mutually exclusive), NPV=16.803.
d) Accept A and D (indivisible, budget 50), max NPV=15.827.
e) Accept A, D, B, 0.533 of E (t1 rationing 15), NPV=19.677.

---The End---